CHENG Zheng

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Research Interests: Construction Grammar, Psycholinguistics, Language Acquisition, Language Comprehension,

EDUCATION

2024.09-2027.06 **Sichuan International Studies University** **Master of Language Intelligence**

GPA：4.72/5.0 (ranking top 5%)

Coursework: Psycholinguistics, Fundamentals of Intelligent Language, Adult Language Cognitive Disorder, Statistics in Language Intelligence, Natural Language Processing, Construction Grammar, Corpus Linguistics,

2020.09-2024.06 **Hubei University of Science and Technology** **Bachelor of English Translation and Interpretation**

GPA：3.25/4.0 (ranking top 5%)

Coursework: Introduction to Linguistics, Introduction to Translation, Fundamental Interpreting, Liaison Interpreting.

Research Experience

Core Member, *FinLingo: AI-Driven Multilingual Financial Text Analysis Workbench* **Jun. 2026 - Aug. 2026**

28th China Robot and AI Competition (Chongqing) | 3rd Prize, AI Innovation

- Linguistically-Driven Knowledge Injection: Built a terminology knowledge base to inject cognitive priors into LLMs. Grounded in Frame Semantics and Register Theory, mapped semantic slots and register-specific definitions into structured fields, resolving cross-lingual ambiguity in low-resource languages.
- Multi-Task SFT & Cross-Lingual Deployment: Fine-tuned 5 financial tasks via LLaMA-Factory, treating prompts as semantic frames to minimize distribution drift. Engineered a single-GPU multi-LoRA hot-switching mechanism, enabling local, multi-lingual processing without external APIs.

Research Assistant, *Computational Stylometry Project* **Jan, 2026 – May. 2026**

Supervised by Prof. Shuqin Jiang, Sichuan International Studies University

- Developed a computational stylometric pipeline analyzing Chinese literary translations, integrating lexical diversity, clause length, syntactic complexity, and readability metrics.
- Engineered a bespoke 14-dimensional syntactic-complexity framework, automating feature extraction to fill a measurable methodological gap where existing NLP tools fell short.

Research Assistant **Aug, 2025 – Present**

Research Group of Prof. Yu (Adam) Deng, Sichuan International Studies University

- Contributed to the full research cycle in construction grammar, event cognition, and language pathology by synthesizing literature, designing materials, and refining protocols.
- Coordinated recruitment and data preprocessing for behavioral and EEG experiments across adult, child, and elderly populations, performing preliminary R-based statistical analyses to feed investigators' models.
- Supported AI-driven mining of discourse data from clinical populations, bridging language intelligence methods with clinical linguistics.
- Streamlined project management and academic dissemination (documentation, seminars, and training) to ensure timely study progression.

Lead Researcher, *Structural Priming in Mandarin Constructions* **Feb. 2025 – Dec. 2025**

Sichuan International Studies University

- Led the full study lifecycle: designed a maze self-paced reading experiment and recruited and ran 160 participants to measure online processing of Mandarin *ba*-constructions.
- Conducted quantitative analyses in R with linear mixed-effects models (LMMs), identifying interaction effects between construction types that support the constructional-network hypothesis.

Research Assistant, Project: *Cross-linguistic Variability in Event Encoding* **Feb. 2025 – Jun. 2025**

Supervisor: Dr. Alimujiang Tusun, University of Cambridge | Faculty of Asian and Middle Eastern Studies

- Executed a behavioral study on language–cognition relations, screening and recruiting 30 monolingual Mandarin speakers to control for linguistic background.
- Solely delivered all 30 video-recorded production sessions (~30 min each), producing 15 hours of primary verbal data for team analysis.

Trainee, Project: ***Eye-Tracking in Translation Studies Summer School*** **Aug. 2025**

South China University of Technology | Guangdong Province Graduate Summer School

- Acquired hands-on experience operating eye-tracking apparatus and designing experimental setups through an intensive one-week program focused on language process research.
- Developed a foundational understanding of utilizing eye-tracking metrics to investigate cognitive and linguistic processing during translation tasks.

Publications & Academic Output

Conference Presentations

- **Cheng, Z.** (2025, May 10-11). [*Does a Caused-Motion Construction Facilitate a Resultative? A Structural Priming Study*]. Oral presentation at the 7th International Conference on (Talmyan) Cognitive Semantics, Kaifeng, China.

Conference Posters

- **Cheng, Z.**, & Deng, Y. (2025, December 12-14). *Does a Caused-Motion Construction Facilitate a Resultative? Evidence from a Structural Priming Study within the Mandarin ba-Construction.* Poster presented at the 19th International Conference on East Asian Psycholinguistics (ICPEAL), Guangzhou, China.
- **Cheng, Z.**, & Deng, Y. (2026, July). *Prototype Effects and Gradience in the Constructional Network: Structural Priming Evidence from Caused-Motion and Resultative Sentences in Mandarin ba-Construction*. Poster presented at the 14th International Conference on Construction Grammar (ICCG-14), Princeton, NJ, USA.

Manuscripts

- **Cheng, Z.**, & Deng, Y. (Under Review). *Prototype Effects and Gradience in the Constructional Network: Structural Priming Evidence from Caused-Motion and Resultative Sentences in Mandarin ba-Construction.* **Review of Cognitive Linguistics.**
- Yang, S., & **Cheng, Z**. (Under Review). *The entrenchment-and-conventionalization mechanism of near-synonymous construction divergence: A collostructional and dynamic semantic space analysis of Chabuduo V and Chadian V*. **Cognitive Linguistics.**
- **Cheng, Z.**, Yang, Y., & Deng, Y. (Under Review). *Continuity and Gradience in the Mandarin ba-Constructional Network: Multivariate and BERT-Assisted Distributional Semantic Analysis of Caused-Motion and Resultative Constructions.* **Language and Cognition.**
- **Cheng, Z.**, Luo, X., & Deng, Y. (Under Review). *From Victim Narratives to Radicalization: A Critical Discourse Analysis of Incel Online Discourse Using Deep Learning*. **Humanities and Social Sciences Communications.**

Dissertation

Master's Thesis

Investigating the Cognitive Mechanisms of the Chinese ba-Construction Network: Converging Evidence from Syntactic Priming and Distributional Semantics Analysis of Caused-Motion and Resultative Constructions

Sichuan International Studies University | Supervised by Prof. Yu (Adam) Deng

- A multi-method thesis validating the constructional network hypothesis for Mandarin, converging behavioral and computational evidence on caused-motion and resultative constructions.
- Study 1 (Behavioral): Designed and executed a syntactic priming experiment using a maze task, collecting and analyzing reaction time data from 160 participants. Findings provide online processing evidence for the network. (This study constitutes a manuscript currently under review at *Review of Cognitive Linguistics*)
- Study 2 (Computational): Embedding-based (BERT) analysis of target constructions on a balanced corpus, quantifying their semantic–syntactic proximity as distributional evidence complementary to Study 1. (A first-author manuscript now under review at *Language and Cognition*)

Skills

Technical & Research Skills

- **Programming & Analysis:** Proficient in R and Python for statistical modeling and computational linguistics. Experienced in designing behavioral experiments with PsychoPy and Ibex Farm.
- **Machine Learning:** Skilled in fine-tuning pre-trained models (e.g., BERT) and applying semi-supervised/unsupervised learning methods to NLP research.
- **Methodologies:** Familiar with experimental psycholinguistics data collection protocols and quantitative corpus-based research methods.

Honors & Awards

Professional Qualifications

- **Test for English Majors (TEM):** Band 4 (Good) & Band 8 (Good)
- **CATTI (China Accreditation Test for Translators and Interpreters):** Level III Translation Certificate

Academic Honors

- **Graduate First-Class Scholarship,** Sichuan International Studies University (Amount/Range, e.g., top 5%)
- **Graduate Thesis Competition, Third Prize, Sichuan International Studies University**
  - **Thesis:** A Comparative Study of Linguistic Features in Depression and Psychosis from the Perspective of Mental State Words and Image Schemas